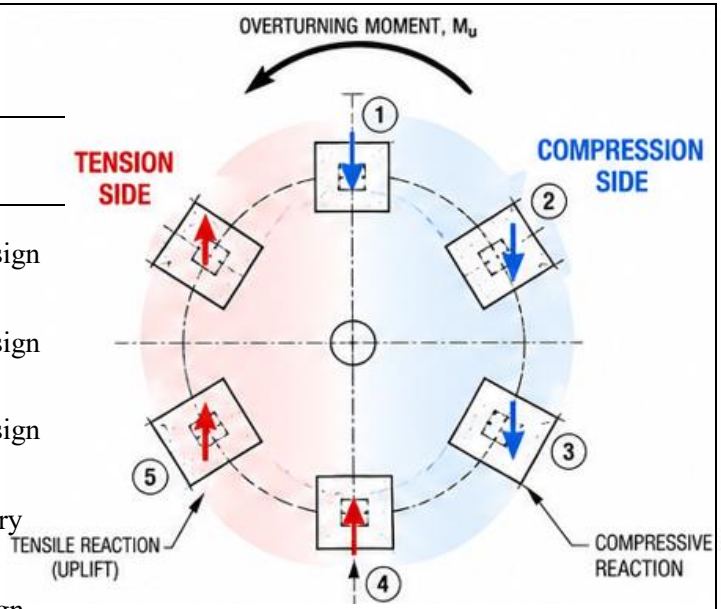
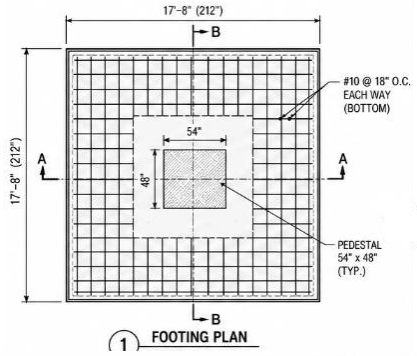
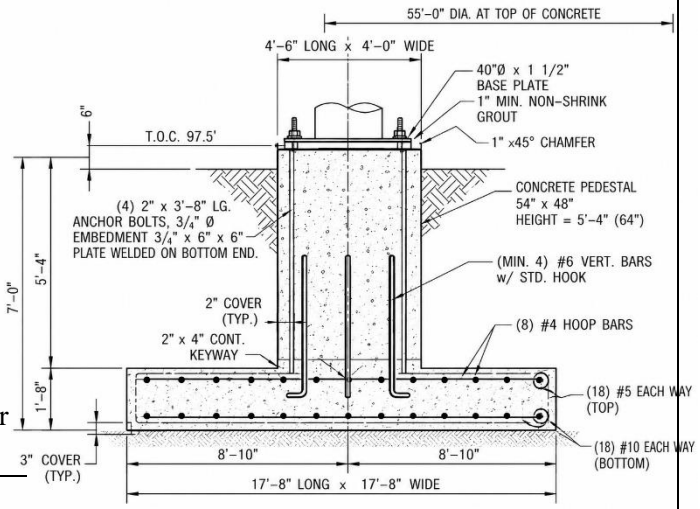
4. Footing segment

No	Design Parameter	Symbol	Value	Source
1	Total factored axial load	P_u	7550.979kips	Main Tank Design
2	Factored overturning moment	M_u	13656.1295kips-ft	Main Tank Design
3	Factored shear	V_u	115.783 kips	Main Tank Design
4	Number of legs	N_{leg}	6	Tank Geometry
5	Pedestal size	$B_p \times L_p$	54 × 48 in	Pedestal Design
6	Pedestal height	H_p	5'-4" = 64 in	Pedestal Design
7	Concrete strength	f'_c	4 ksi	User Input
8	Reinforcing steel	f_y	60 ksi	User Input
9	Concrete unit weight	γ_c	150 pcf	From ACI Code
10 11	Assumed footing thickness	h	30 in	Assumed
12	Concrete cover	c_c	3 in	Assumed
13	Bottom reinforcement		#10 bars, each way	User input
14	Allowable soil pressure (Assumed)	q_{allow}	3.0 ksf	User Input
15	Footing dimensions	$B \times L$	17′-8″ × 17′-8″	From drawing (user input)







Step 1: Footing Load

Gravity load per leg

$$P_{\text{leg}} = \frac{P_u}{N_{\text{leg}}}$$

$$P_{leg} = \frac{7550.979}{6} = 1258.4965 \text{ kip}$$

Self-weight of Pedestal

Factored self-weight, assuming a 1.2 dead-load factor:

$$W_{p,u} = 1.2(14.40) = 17.28 \text{ kips}$$

$$\boldsymbol{P_u = 1275.777 \text{ kips}}$$

Step 2: Preliminary Footing Size

Required footing area

Footing Area:

$$A_{req} = \frac{P_u}{q_{allow}} = \frac{1275.77}{3} = 425.26\ ft^2$$

$$A_{req} = \sqrt{A_{req}} = \sqrt{425.26} = 20.62 \text{ ft}$$

Specified trial dimensions: B = L = 17 + 8/12 = 17.6667 ft.

$$B = L = 17.6667 \text{ ft}$$

$$A_f = 17.6667(17.6667)$$

$$\boldsymbol{A_f = 312.11 \text{ ft}^2}$$

Note: This sizing uses a factored load from the drawing, the designer should use required sizing for the design of the footing.

Step 3: Soil Bearing Check

Average factored bearing pressure:

$$q_u = \frac{P_u}{A_f}$$

$$q = \frac{1275.776}{312.111}$$

$$q = 4.088 \text{ ksf}$$

Note: This factored pressure (4.088 ksf) must not be compared directly with qallow = 3.0 ksf.

A valid allowable-bearing check requires service loads and the moment-induced pressure variation.

Concrete exposure	Member	Reinforcement	Specified cover, mm
Cast against and permanently in contact with ground	All	All	75
Exposed to weather or in contact with ground	All	No. 19 through No. 57 bars	50
		No. 16 bar, MW200 or MD200 wire, and smaller	40
Not exposed to weather or in contact with ground	Slabs, joists and walls	No. 43 and No. 57 bars	40
		No. 36 bar and smaller	20
	Beams, columns, pedestals and tension ties	Primary reinforcement, stirrups, ties, spirals and hoops	40

Table 20.6.1.3.1 Specified concrete cover for cast-in-place nonprestressed concrete members

Step 4: Footing Thickness

Adopt:

$$h_{trail} = 30 \text{ in.}$$

Note: The software shall subsequently perform the required structural checks using the selected footing thickness. If any required check does not satisfy the applicable ACI 318 provisions, the software shall display a **warning/red alert** and the user shall increase the footing thickness $h$and repeat the checks until all required criteria are satisfied. (Recommended 25% - 30%h_{trail}).

Given:

$$B = L = 17.6667 \text{ ft}$$

$$f'_c = 4.0 \text{ ksi}$$

Bottom reinforcement (User Input) $\rho_{b_1} = \rho_{b_2}$:

#8 @ s each way

Concrete cover: Concrete cover shall be determined in accordance with the applicable **ACI 318 requirements table 20.6.1.3.1** for the footing exposure and construction condition.

3 in.

For #10 bars use Table A-1:

$d_b = 1.27$ in.

Therefore:

$$d = h - \text{cover} - \frac{d_b}{2}$$

$$d = 30 - 3 - \frac{1.27}{2}$$

ASTM Standard Reinforcing Bar — Nominal Diameter

Bar designation	Nominal diameter, d_b (in.)
#3	0.375
#4	0.500
#5	0.625
#6	0.750
#7	0.875
#8	**1.000**
#9	1.128
#10	1.270
#11	1.410
#14	1.693
#18	2.257

Source: ACI 318-14, Appendix A—Steel Reinforcement Information; ASTM standard reinforcing bars.

$$d = 26.365 \text{ in} = 2.1971 \text{ft}$$

Step 5: Two-Way Shear / Punching Shear

[ACI 318-19: 22.6.4.1, 22.6.5.2 & 3]

For a square pedestal, the critical punching-shear perimeter is located approximately $\frac{d}{2}$ from the pedestal Face, (Pedestal dimensions: c1 = 54 in.; c2 = 48 in).

$$d = 26.365 \; in.$$

$$b_o = 2(c1 + c2 + 2d)$$

$$b_o = 2(54 + 48 + 2(26.365)) = 309.46 \text{ in}$$

or:

$$b_o = \frac{309.46}{12} = 25.7883 \text{ ft}$$

Area inside the critical perimeter:

$$A_o = (c1 + d)(c2 + d)$$

$$A_o = (54 + 26.365)(48 + 26.365)$$

$$A_o = 5976.3 \text{ in}^2 = 41.50 \text{ ft}^2$$

Soil reaction inside:

$$P_{soil} = q_u A_o$$

$$P_{soil} = 4.088(41.50)$$

$$P_{soil} = 169.64 \text{ kip}$$

Punching shear demand:

$$V_u = P_{u,\text{leg}} - P_{\text{soil}}$$

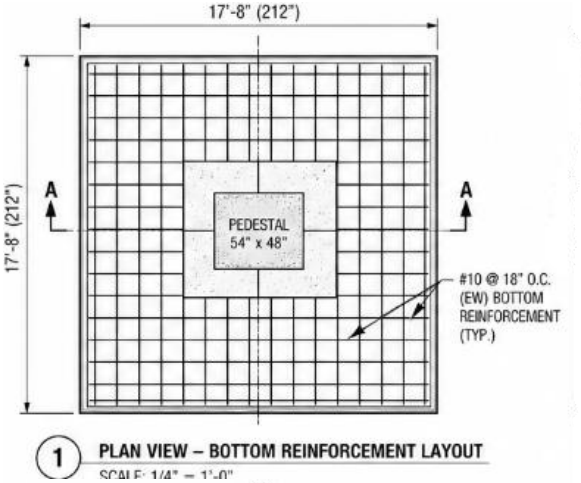
$$V_u = 1275.776 - 169.64$$

$$1106.13 \text{ kips}$$

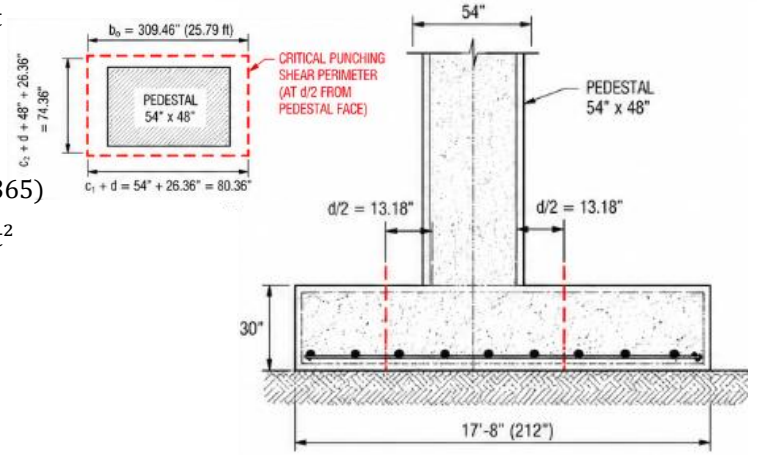
Therefore:

$$\beta = \frac{\max(c1, c2)}{\min(c1, c2)} = 1.125$$

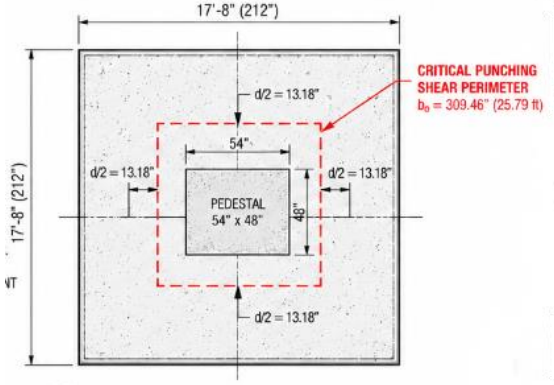
So $\beta = 1$means the loaded area is square (The design is limited to square cross-section).



1 PLAN VIEW – BOTTOM REINFORCEMENT LAYOUT



5 SECTION - PUNCHING SHEAR
SCALE: 1/2" = 1'-0"



2 PLAN VIEW – PEDESTAL & PUNCHING SHEAR
SCALE: 1/4" = 1'-0"

$$\beta = 1.125$$

location of pedestal

$\alpha_s$depends on where the loaded area is located (ACI 318-19 22.6.5.3):

Pedestal location	α_s
Interior	**40**
Edge	**30**
Corner	**20**

Table 19.2.4.1(a)—Values of λ for lightweight concrete based on equilibrium density

w_c, lb/ft³	λ	
≤ 100	0.75	(a)
100 < w_c ≤ 135	0.0075w_c ≤ 1.0	(b)
> 135	1.0	(c)

Your footing is being designed as an interior/square pedestal condition, so:

$\alpha_s = 40$

Equation 1: where $\beta \leq 2$

$$\phi V_c = 0.75\left(4\lambda_s\lambda\sqrt{f_c'}\right)b_o d$$

However, for **isolated footings**, the size-effect factor may be taken as:

$$\lambda = 1.0$$

The size-effect factor may be neglected for shallow foundations based ACI-318-19.

Thus:

$$\phi V_c = 0.75(4)(1)(1)\sqrt{4000}(309.46)(26.365)/1000$$

$$\phi V_{c1} = 1548.04\ kip$$

The other two expressions give approximately:

Equation 3

$$\phi V_c = 0.75\left(2 + \frac{\alpha_s d}{b_o}\right)\lambda_s\lambda\sqrt{f'_c}\,b_o d$$

$$\phi V_{c3} = \frac{0.75\left(2 + \frac{40(26.365)}{309.46}\right)(1)(1)\sqrt{4000}(309.46)(26.365)}{1000}$$

$$\phi V_{c3} = 2092.91\ kip$$

$$V_c = \min(1548.04, 2092.91)$$

Therefore:

$$\phi V_c = 1548.04\ kip$$

Note: Correction to your original procedure: The equation for V_{c3} must use the factor $(2 + \alpha_s d/b_o)$ without dividing that factor by 12. The same principle applies to the other two expressions. The factor of 1,000 in the calculations above is only for converting pounds to kips.

Check:

$$V_u = 1106.14 \text{ kips}$$

The design strength is:

$$\phi V_c = 1548.04 \text{ kips}$$

Check:

$$V_u \leq \phi V_c$$

Utilization:

$$1106.14 \leq 1548.04$$

$$\text{Utilization} = \frac{V_u}{\phi V_c} \times 100\%$$

$$= \frac{1106.14}{1548.04} \times 100\%$$

$$\textbf{Utilization} = \mathbf{71.45}\%$$

Step 7: One-Way Shear

The footing projection from the pedestal face to the footing edge is:

$$l_1 = \frac{B - 54/12}{2} = \frac{17.667 - 4.50}{2} = 6.583 \; ft$$

$$l_2 = \frac{L - 48/12}{2} = \frac{17.667 - 4.00}{2} = 6.833 \; ft$$

For a 1-ft strip:

$$V_{u,1} = q_u(l_1 - d) = \; 4.088(6.583 - 2.1971) = 17.93 \; kips/ft$$

For a 1-ft strip:

$$V_{u,2} = q_u(l_2 - d) = \; 4.088(6.383 - 2.1971) = 18.95 kips/ft$$

For normal-weight concrete, $\lambda = 1.0$, and:

$$V_c = 2\lambda\sqrt{f_c'} b_w d$$

$$V_c = 2(1.0)\sqrt{4000}(12)(26.365)/1000$$

$$V_c = 40.02 \; kip$$

With $\phi = 0.75$:

$$\phi V_c = 0.75(40.02)$$

$$\phi V_c = 30.01\ kip$$

Check:

$Direction\ 1: 17.93\ <\ 30.01\ kips/ft\ -\ preliminary\ PASS\ (utilization\ 59.7\%)$.

$Direction\ 2: 18.95\ <\ 30.01 \frac{kips}{ft} -\ preliminary\ PASS\ (utilization\ 63.1\%)$.

Step 8: Flexure and Reinforcement

For a 12-in.-wide strip:

$$b = 12\ in.$$

$$d = 26.365\ in$$

$$f_c' = 4.0\ ksi$$

$$f_y = 60\ ksi$$

$$\phi = 0.90$$

For a uniformly distributed factored soil pressure:

$$M_u = \frac{ql^2}{2}$$

$$M_{u,1} = \frac{4.088(6.583)^2}{2}$$

$$M_u = 88.58\ kip - ft/ft$$

$$\boldsymbol{M_u = 1062.9\ kip - in./ft}$$

$$M_{u,2} = \frac{4.088(6.833)^2}{2}$$

$$M_u = 95.43\ kip - ft/ft$$

$$\boldsymbol{M_u = 1145.2\ kip - in./ft}$$

the nominal moment strength is:

$$M_n = A_s f_y \left(d - \frac{a}{2}\right)$$

where

$$a = \frac{A_s f_y}{0.85 f_c' b}$$

Therefore:

$$\phi A_s f_y \left[d - \frac{A_s f_y}{2(0.85 f_c' b)} \right] = M_u$$

Substituting:

$$0.90 A_s (60) \left[26.365 - \frac{A_s (60)}{2(0.85)(4)(12)} \right] = 358.16$$

Solving for A_s:

$$A_{s,req} = \frac{0.251\ in^2}{ft}$$

Minimum reinforcement

$$A_{s,min} = \rho_{min} b h$$

Using:

$$\rho_{min} = 0.0018$$
$$A_{s,min} = (0.0018)(12)(30)$$
$$A_{s,min} = \frac{0.648\ in^2}{ft}$$

Therefore, the governing required reinforcement is:

$$A_{s,design} = \max(A_{s,req}, A_{s,min})$$
$$A_{s,design} = \max(0.252, 0.648)$$
$$A_{s,design} = \frac{0.648\ in^2}{ft}$$

Provided reinforcement

Trial #10 @ 18 in. each way provides As = 1.27 × 12/18 = 0.847 in²/ft.

The reinforcement provided per foot is:

$$A_{s,prov} = A_b \frac{12}{s}$$

$$A_{s,prov} = 1.27(\frac{12}{18})$$

$$A_{s,prov} = \frac{0.847\ in^2}{ft}$$

For the provided steel: a = 1.245 in.; φMn = 1176.9 kip-in./ft.

Check against required reinforcement:

$$0.677 > 0.648$$

$$A_{s,prov} \geq A_{s,design}\ PASS$$

Flexural strength check

Flexure comparison:

Direction 1 φMn 1176.9 vs Mu1 1062.9 kip-in./ft

Direction 2 φMn 1176.9 vs Mu2 1145.2 kip-in./ft

FLEXURE: PASS

Step 9: Pedestal/Footing Bearing

$$\varphi Pn\ = \phi(0.85 f_c' A_1)$$

with:

$$\phi = 0.65$$

For your 54 × 48 pedestal:

$$A_1 = 54 \times 48 = 2592\ in^2.$$

Therefore:

$$\varphi Pn = 0.65(0.85)(4)2592 = 5728.3\ kips.$$

Compare:

$$P_u = 520\ kip$$

$$1275.7765 < 5728.32\ kips \ \ldots\ldots\ldots \textit{Concrete Bearing — PASS}$$

Step 10: Dowel / Pedestal-to-Footing Reinforcement

For a cast-in-place pedestal connected to a foundation, ACI 318-14 $16.3.4.1 states:

$$A_{s,\text{dowel}} \geq 0.005 A_g$$

where A_g is the gross cross-sectional area of the supported pedestal/column and at least four dowels.

Based on pedestal:

$$A_{s,min} = 0.005(\ 2592\ )$$

$$A_{s,min} = 12.96\ \ in^2$$

24 #6 bars would provide 24 × 0.44 = 10.56 in², exceeding that assumed area criterion.

Therefore:

12.96 $>$ *10.56..... DOWEL/INTERFACE STEEL — PASS*

As long as the assumed size is not sufficient, we should choose the minimum As.

Step 11: Development of deformed bars in compression

$$l_{dc} = \frac{f_y \psi_r}{50\lambda\sqrt{f_c'}} d_b$$

with the minimum requirement:

$$0.0003 f_y d_b \geq 8\ in.$$

For #6:

$$d_b = 0.75\ in.$$

$$l_{dc} = \frac{60{,}000(1.0)}{50(1)\left(\sqrt{4000}\right)}(0.75)$$

$l_{dc} \approx 14.23$ in.

Minimum:

$$0.0003(60,000)(0.75) = 13.5\ in.$$

Therefore:

$$l_{dc} = 14.23\ in$$

For detailing, we can adopt:

$$18 > 14.23$$

Item	Result	Status / note
Trial footing plan dimensions	17 ft 8 in. × 17 ft 8 in. (312.11 ft²)	Specified; square
Preliminary per-leg factored load including pedestal	1275.78 kips	Assumes total axial load shared equally among 6 legs
Average factored pressure	4.088 ksf	Not comparable directly with allowable 3.0 ksf
Allowable bearing including overturning	Not verified	Service loads and moment distribution required
Punching shear	1106.1 kips demand vs 174.4 kips strength	FAIL under preliminary assumptions
One-way shear, direction 1	17.93 vs 30.01 kips/ft	Preliminary pass
One-way shear, direction 2	18.95 vs 30.01 kips/ft	Preliminary pass
Bottom reinforcement trial	#10 @ 18 in. each way; As = 0.847 in²/ft	Preliminary flexure only; spacing not supplied in input
Pedestal bearing	1275.8 vs 5728.3 kips	Concentric preliminary check